

CIE

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TITLE: Nix - AI Coding Assistant

SCHOOL: Chitkara University, Rajpura, Punjab

**DETAIL OF THE STUDENT/MENTOR WHO UPLOADED THE COPYRIGHT ON GOVT PORTAL NEED TO
SHARE LOGIN AND PASSWORD WITH MENTORS**

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ABSTRACT

Nix is an AI-based command-line wrapper that assists developers to analyze, comprehend, and debug Spring Boot apps via a chat terminal. The platform resolves a major sore point of developing a backend - the contrast between looking at unfamiliar code and knowing what it does, by embedding context-specific AI help into the workspace of the developer, as well as not needing extra software or even documentation search.

The fundamental aspect of Nix is that it allows developers to index current Spring Boot codebase and query them with natural language interaction in the terminal. Among the most important highlights of the platform is that the whole platform runs on the local machine of the user, and thus sensitive or proprietary code does not leave the environment of the developer. Local-first This is about the codebase privacy that you can achieve and at the same time offer fast and accurate responses in Groq LLM API without needing a subscription-based service.

Nix also offers structured understanding of Spring Boot-specific components, like bean dependencies, REST endpoint mapping, JPA entity relationship, and inter-service call paths to enhance both the ease with which code is understood and the amount of effort required to trace code. The platform uses AST Tree-sitter to analyze source code and operates based on Java AST analysis to fulfill responses based on the project structure, and not generalized assumptions. This renders Nix especially useful in onboarding to new codebases or rearing out intricate dependency chains.

It also has a conversational agent implementation, which remembers query context between interactions to enable developers to ask follow-up questions and dynamically navigate their codebase. The system now offers twenty four Spring aware tools, including dependency, endpoint discovery, Spring Bean inspection and call chain tracing. The tools are constructed over Spring Boot conventions instead of generic code search and generate structurally correct responses instead of text matches.

Nix, using a mix of local codebase indexing, AST-based code parsing, Spring-aware tooling, conversational querying, and a lightweight set up process, develops a pragmatic and refined developer helper in the Spring Boot ecosystem. It can save time on circular debugging processes, faster onboarding onto substantial codebases, and provide developers with a dependable method to investigate and comprehend their projects, making it an effective instrument in contemporary Java back-end development processes.

1.1 Problem Background

The developers of Spring boot who have to deal with large or unknown codebase take some time before they can debug anything meaningful or extend anything, to figure out the bean dependencies, endpoint mappings and service interactions manually. Current tools such as IDEs and documentation provide partial assistance, but constantly demand context switching. No local, dedicated assistant, which can be run locally that is aware of the Spring Boot conventions and allows developers to talk to their own codebase in a natural language, without leakage of proprietary code to third-party services.

1.2 Objective of the Project

To create an AI-assisted local command-line helper that allows the developers of Spring Boot to query their codebase using natural language and lessen the amount of time spent by hand tracing code, and enhance the efficiency of the debugging process without reducing the privacy of the source code.

1.3 Proposed Solution / System Overview

Nix is a terminal-based artificial intelligence agent that uses Tree-sitter to scan Spring Boot projects and analyze Java AST-based data, followed by a structured local index and answers questions asked by developers using a conversational language agent based on the Groq LLM API. These system major components include codebase indexer, a spring-aware tool set, a retrieval engine, and an agent layer that can interactive layer that together with the specific structure of the project developed by the developer can give a correct and context-based response and is not based on general knowledge base.

1.4 Key Features or Functionality

Nix offers twenty-four tools that are Spring-aware and include bean dependency analysis, REST endpoint discovery, JPA entity mapping and call chain tracing. The developers communicate in a chat style interface that allows follow-up questions. The system is entirely locally operated, and source code confidential. Answers are based on AST-scanned project structure, not general assumptions and are structurally correct to the particular codebase being asked.

1.5 Technology / Method Used

Nix is implemented in Python with Tree-sitter to do Java AST parsing and a local JSON index of extracted structural metadata. The Groq LLM API offers inference and a conversational agent architecture coordinates tool choice and multi-turn query context throughout interactions.

1.6 Expected Outcome & Benefits

Nix will dramatically decrease the time developers take to understand new Spring Boot projects. It helps with debugging and onboarding by providing the bean's hierarchy, the API's structure and the call graph as needed. Its local-first approach does not rely on exposing code, so it's suitable for enterprise development. Developers have an accessible, persistent sidekick within their terminal, without needing to learn new tools.

1.7 Original Contribution Statement

Nix adds a locally-executable, Spring Boot-specific conversational code helper, integrating AST-based structural parsing and LLM-based natural language querying - designed to integrate into Java development workflows based on Java backend development.

RESULT:-

```
nikhilsaini@Nikhils-MacBook-Air demo % nix

                                     NIX
                                     AI-powered assistant for Spring Boot projects

Indexing 3 Java files.../Users/nikhilsaini/.local/pipx/venvs/nix/lib/python3.14/site-packages/chromadb/config.py:20: UserWarning: Core Pydantic V1 functionality isn't compatible with Py
hon 3.14 or greater.
  from pydantic.v1 import BaseSettings
✓ Spring Boot 4.0.2 • 3 files indexed

Type your question or 'help' for options. Ctrl+C to exit.

> help
Commands:
help      Show this help
status    Show project status
new       Start fresh conversation
clear     Clear the screen
exit      Exit nix

Or just type naturally:
analyze my dependencies
analyze everything
what is my project structure

> what can you do nix

You are Nix, an AI-powered CLI assistant for Spring Boot projects. I was created by Nikhil (github.com/Nikhilsaini2204). My capabilities include analyzing dependencies, finding issues/bugs
, diagnosing errors, searching code, analyzing endpoints/services/entities, explaining configuration, and more.

> can you tell me how many dependencies in the project

Using 1 tool...
• analyze_dependencies
  4 dependencies (4 Spring, 2 test)
  └─ spring-boot-starter-webmvc
      └─ compileOnly 'org.projectlombok:lombok'
  └─ spring-boot-starter-webmvc-test
      └─ testRuntimeOnly 'org.junit.platform:junit-platform-launcher'
  └─ spring-boot-starter-webmvc:managed
  └─ spring-boot-starter-webmvc-test:managed

Found 4 dependencies (4 Spring-related, 2 test dependencies)

> can you tell me how many controllers in the project
```